

HIMANSHU DHURVE

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SUMMARY

ML engineer with hands-on experience in post-training alignment, model evaluation, and deployment pipelines. I am drawn to the gap between a model that performs well under evaluation and one whose internal behaviour can actually be trusted, and increasingly convinced that closing that gap is the most important engineering problem in AI right now. Looking to contribute to technical safety research grounded in real systems: interpretability-informed evaluation, runtime monitoring, and the kind of rigorous measurement methodology that distinguishes genuine alignment from performed alignment.

EDUCATION

San Diego State University

Master of Science in Aerospace Engineering, GPA: 3.52/4.0

San Diego, USA

2021 – 2024

Dr. Babasaheb Ambedkar Technological University

Bachelor of Technology in Mechanical Engineering, GPA: 8.57/10.0

Raigad, India

2014 – 2018

PROFESSIONAL EXPERIENCE

AI/ML Research Intern, Vizura AI Labs, India

Jan 2026 – March 2026

- Designed and deployed a **multimodal RAG evaluation framework** for technical documentation (OpenFOAM); introduced a **six-dimension LLM-as-a-judge benchmark** across groundedness, citation correctness, factual recall, and coverage, achieving overall scores **4.6+/5.0**; this **document processing pipeline** is directly applicable to form extraction and structured output retrieval from technical corpora. Submitted to *RAG4Rep Workshop, ACL 2025*. [\[openfoam-ai-guide\]](#)
- Developed a **signal processing pipeline** for SAR (Synthetic Aperture Radar) image analysis: fine-tuned **RF-DETR** (transformer-based detector) on a custom multi-class dataset achieving mAP **0.614**.
- Built an **agentic vision-language model (VLM) pipeline** converting research PDFs into structured digests by autonomously generating narrative summaries, AI-illustrated diagrams, and executable Jupyter notebooks via coordinated **VLM reasoning**.

Aerodynamics Research Assistant, San Diego State University, USA

Aug 2021 – Dec 2023

- Designed **large-scale data acquisition and analysis pipelines** on an HPC cluster (SLURM); processed high-dimensionality flow field datasets in Python, cutting simulation wall-time by **75%** through parallelization.

TECHNICAL SKILLS

Core Competencies	Vision-Language Models (VLMs), LLM Evaluation & Benchmarking, Post-Training Alignment
Languages & Tools	Python, C++, PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, Git
MLOps & Infra	Databricks, MLFlow, DVC, Docker, CI/CD, OpenAI SDK, FastAPI, uv
Research Methods	Dataset Curation, Benchmark Design, Human-in-the-Loop Assessment, Model Evaluation, Quantization

PROJECTS

End-to-End MLOps Pipeline for Customer Churn Prediction

Apr 2026

- Built a fully modular production pipeline on 243k rows: **XGBoost** with **Optuna** tuning (30 trials, recall-optimised), **MLflow** experiment tracking and model registry, **FastAPI + Gradio** serving, **multi-stage Docker** build, and **GitHub Actions** CI/CD, achieving **93% recall** at a tuned threshold of 0.30 on an imbalanced churn dataset. [\[repo\]](#)

LLM Evaluation via Multi-Signal Reward Engineering (Medical MCQ)

Nov 2025

- Fine-tuned **Qwen3-1.7B** with **GRPO** on structured medical QA using **multi-signal reward functions** (binary correctness, regex format validation, length-calibrated explanation scoring); managed full **SFT + GRPO pipeline** on constrained Kaggle GPUs with 4-bit quantization (LoRA rank 32) and **vLLM-accelerated sampling**, analysed failure modes and output distributions to identify model weaknesses. [\[repo\]](#)

Temporal Signal Stabilization for Video OCR (License Plate Recognition)

Sep 2025

- Built a real-time detection and recognition pipeline using fine-tuned **YOLO11** and **EasyOCR**; addressed **temporal signal noise** (OCR flickering) with a majority-voting algorithm over a 12-frame rolling buffer. [\[repo\]](#)

Video Instance Segmentation for Scene Analysis

Sep 2025

- Implemented per-frame object detection and pixel-level segmentation using PyTorch's pre-trained **Mask R-CNN** on video streams, producing per-instance coloured masks for audio-visual scene understanding and spatial activity localisation. [\[repo\]](#)

Multimodal Benchmarking, Student Misconception Detection (Kaggle MAP)

Aug – Oct 2025

- Fine-tuned **Qwen2.5-Math-1.5B-Instruct** with **Unsloth** and a compact **MLP classification head** on curated math reasoning misconception benchmarks; improved MAP leaderboard score from 0.885 to **0.925** (+4.5%) by combining LLM semantic understanding with structured output evaluation. [\[repo\]](#)

AI SAFETY ENGAGEMENT

- **BlueDot Impact**, Technical AI Safety (Intensive Cohort, May 2026)

PUBLICATIONS

- **Himanshu C. Dhurve** et al. “Decompose, Retrieve, Cite: A RAG Pipeline for Structured Report Generation from Technical Documentation.” *RAG4Report Workshop, ACL 2026* (First Author).
- Joseph Katz, **Himanshu C. Dhurve**. “Effect of Surface Texture on the Lift and Drag of Small Spinning Balls.” *AIAA Aviation Forum and Ascend 2025*. [\[Paper Link\]](#)

HONORS & CERTIFICATIONS

- Machine Learning Specialization, Stanford University & DeepLearning.AI

Jan 2025
- Recipient of San Diego State University Master’s Research Scholarship

May 2022

Public Repositories: [GitHub](#)

Publications: [AIAA](#)